

ANALYTICAL REPORT

TestAmerica Laboratories, Inc.

TestAmerica Michigan

10448 Citation Drive

Suite 200

Brighton, MI 48116

Tel: (810)229-2763

TestAmerica Job ID: 190-18036-1

Client Project/Site: IPP - Avon

For:

City of Cadillac Utilities

1121 Plett Road

Cadillac, Michigan 49601

Attn: Cindy Tomaszewski



Authorized for release by:

12/10/2018 6:26:09 PM

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Results relate only to the items tested and the sample(s) as received by the laboratory.



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: IPP - Avon

TestAmerica Job ID: 190-18036-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-18036-1	Field Blank	Water	11/07/18 11:10	11/08/18 12:41
190-18036-2	Avon Auto	Water	11/07/18 11:15	11/08/18 12:41
190-18036-3	Avon 5th Street	Water	11/07/18 11:25	11/08/18 12:41
190-18036-4	Avon Duplic.	Water	11/07/18 00:00	11/08/18 12:41

Case Narrative

Client: City of Cadillac Utilities
Project/Site: IPP - Avon

TestAmerica Job ID: 190-18036-1

Job ID: 190-18036-1

Laboratory: TestAmerica Michigan

Narrative

Job Narrative 190-18036-1

Comments

The PFAS analysis was performed at Eurofins Lancaster Laboratories.

Receipt

The samples were received on 11/8/2018 12:41 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 1.2° C.

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.



ANALYSIS REPORT

Prepared by:

Eurofins Lancaster Laboratories Environmental
2425 New Holland Pike
Lancaster, PA 17601

Prepared for:

TestAmerica
10448 Citation Drive
Suite 200
Brighton MI 48116

Report Date: November 29, 2018 15:45

Project: IPP - Avon

Account #: 01042
Group Number: 2010206
PO Number: 19001884
State of Sample Origin: MI

Electronic Copy To TestAmerica

Attn: Sue Schafer

Respectfully Submitted,



Wendy A. Kozma
Principal Specialist Group Leader

(717) 556-7257

To view our laboratory's current scopes of accreditation please go to <http://www.eurofinsus.com/environment-testing/laboratories/eurofins-lancaster-laboratories-environmental/resources/certifications/>. Historical copies may be requested through your project manager.



SAMPLE INFORMATION

<u>Client Sample Description</u>	<u>Sample Collection Date/Time</u>	<u>ELLE#</u>
Field Blank (190-18036-1) Water	11/07/2018 11:10	9903347
Avon Auto (190-18036-2) Water	11/07/2018 11:15	9903348
Avon 5th Street (190-18036-3) Water	11/07/2018 11:25	9903349
Avon Duplic. (190-18036-4) Water	11/07/2018	9903350

The specific methodologies used in obtaining the enclosed analytical results are indicated on the Laboratory Sample Analysis Record.

Sample Description: Field Blank (190-18036-1) Water
IPP-Avon

TestAmerica
ELLE Sample #: WW 9903347
ELLE Group #: 2010206
Matrix: Water

Project Name: IPP - Avon

Submittal Date/Time: 11/16/2018 11:40

Collection Date/Time: 11/07/2018 11:10

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.85	1
14473	6:2 fluorotelomersulfonate	27619-97-2	3.5	0.85	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.7	1
14473	NEtFOSAA	2991-50-6	N.D.	0.85	1
NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	N.D.	0.85	1
NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorononanesulfonate (PfnS)	68259-12-1	N.D.	0.51	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.7	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.26	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.77	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.43	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.51	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.34	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.34	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.34	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.34	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.34	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.26	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.43	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	1.4 J	0.34	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.7	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.34	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.26	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.34	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.34	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 09:41	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

Analysis Report

Sample Description: Avon Auto (190-18036-2) Water
IPP-Avon

TestAmerica
ELLE Sample #: WW 9903348
ELLE Group #: 2010206
Matrix: Water

Project Name: IPP - Avon

Submittal Date/Time: 11/16/2018 11:40

Collection Date/Time: 11/07/2018 11:15

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	1.2	1
14473	6:2 fluorotelomersulfonate	27619-97-2	14	1.2	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	2.5	1
14473	NEtFOSAA	2991-50-6	N.D.	1.2	1
NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	N.D.	1.2	1
NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorononanesulfonate (PfnS)	68259-12-1	N.D.	0.75	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	17	2.5	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	660	3.7	10
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	1.1	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.62	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	16	0.75	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.50	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.50	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.50	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.50	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	1.0 J	0.50	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	0.94 J	0.37	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.62	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	7.1	0.50	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	2.5	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.50	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.37	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.50	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.50	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

Reporting limits were raised due to interference from the sample matrix.

The sample injection internal standard peak areas were outside of the QC limits for both the initial injection and the re-injection. The values here are from the initial injection of the sample.

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 09:50	Jason W Knight	1
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 16:11	Jason W Knight	10

Sample Description: Avon Auto (190-18036-2) Water
IPP-Avon

TestAmerica
ELLE Sample #: WW 9903348
ELLE Group #: 2010206
Matrix: Water

Project Name: IPP - Avon

Submittal Date/Time: 11/16/2018 11:40

Collection Date/Time: 11/07/2018 11:15

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

Sample Description: Avon 5th Street (190-18036-3) Water
IPP-Avon

TestAmerica
ELLE Sample #: WW 9903349
ELLE Group #: 2010206
Matrix: Water

Project Name: IPP - Avon

Submittal Date/Time: 11/16/2018 11:40

Collection Date/Time: 11/07/2018 11:25

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.92	1
14473	6:2 fluorotelomersulfonate	27619-97-2	4.3	0.92	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.8	1
14473	NEtFOSAA	2991-50-6	N.D.	0.92	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.				
14473	NMeFOSAA	2355-31-9	N.D.	0.92	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.				
14473	Perfluorononanesulfonate (PfnS)	68259-12-1	N.D.	0.55	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.8	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.27	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.82	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.46	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.55	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.37	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.37	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.37	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.37	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.37	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.27	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.46	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	1.8	0.37	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.8	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.37	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.27	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.37	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.37	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 09:59	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

Sample Description: Avon Duplic. (190-18036-4) Water
IPP-Avon

TestAmerica
ELLE Sample #: WW 9903350
ELLE Group #: 2010206
Matrix: Water

Project Name: IPP - Avon

Submittal Date/Time: 11/16/2018 11:40

Collection Date/Time: 11/07/2018

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.94	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	0.94	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.9	1
14473	NEtFOSAA	2991-50-6	N.D.	0.94	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.				
14473	NMeFOSAA	2355-31-9	N.D.	0.94	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.				
14473	Perfluorononanesulfonate (PfnS)	68259-12-1	N.D.	0.57	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.9	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.28	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.85	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.47	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.57	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.38	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.38	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.38	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.38	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.38	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.28	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.47	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	N.D.	0.38	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.9	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.38	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.28	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.38	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.38	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 10:08	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

Quality Control Summary

Client Name: TestAmerica
Reported: 11/29/2018 15:45

Group Number: 2010206

Matrix QC may not be reported if insufficient sample or site-specific QC samples were not submitted. In these situations, to demonstrate precision and accuracy at a batch level, a LCS/LCSD was performed, unless otherwise specified in the method.

All Inorganic Initial Calibration and Continuing Calibration Blanks met acceptable method criteria unless otherwise noted on the Analysis Report.

Method Blank

Analysis Name	Result ng/l	MDL ng/l
Batch number: 18321001	Sample number(s): 9903347-9903350	
4:2 fluorotelomersulfonate	N.D.	1.0
6:2 fluorotelomersulfonate	N.D.	1.0
8:2 fluorotelomersulfonate	N.D.	2.0
NEtFOSAA	N.D.	1.0
NMeFOSAA	N.D.	1.0
Perfluorononanesulfonate(Pfns)	N.D.	0.60
Pfba-Perfluorobutanoic Acid	N.D.	2.0
Pfbs-Perfluorobutanesulfonate	N.D.	0.30
Pfda-Perfluorodecanoic Acid	N.D.	0.90
Pfdoda-Perfluorododecanoic	N.D.	0.50
Pfds-Perfluorodecanesulfonate	N.D.	0.60
Pfhpa-Perfluoroheptanoic Acid	N.D.	0.40
Pfhps-Perfluoroheptanesulfonat	N.D.	0.40
Pfhxa-Perfluorohexanoic Acid	N.D.	0.40
Pfhxs-Perfluorohexanesulfonate	N.D.	0.40
Pfna-Perfluorononanoic Acid	N.D.	0.40
Pfoa-Perfluorooctanoic Acid	N.D.	0.30
Pfosa-Perfluorooctanesulfonami	N.D.	0.50
Pfos-Perfluorooctanesulfonate	N.D.	0.40
Pfpea-Perfluoropentanoic Acid	N.D.	2.0
Pfpes-Perfluoropentanesulfonat	N.D.	0.40
Pfteda-Perfluorotetradecanoic	N.D.	0.30
Pftrda-Perfluorotridecanoic Ac	N.D.	0.40
Pfunda-Perfluoroundecanoic Aci	N.D.	0.40

LCS/LCSD

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Batch number: 18321001	Sample number(s): 9903347-9903350								
4:2 fluorotelomersulfonate	14.94	12.84	14.94	13.35	86	89	82-152	4	30
6:2 fluorotelomersulfonate	15.17	13.88	15.17	13.23	92	87	66-155	5	30
8:2 fluorotelomersulfonate	15.33	14.01	15.33	13.74	91	90	66-148	2	30
NEtFOSAA	5.44	5.43	5.44	4.83	100	89	55-169	12	30
NMeFOSAA	5.44	3.86	5.44	4.27	71	78	62-167	10	30

*- Outside of specification

- (1) The result for one or both determinations was less than five times the LOQ.
- (2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica
Reported: 11/29/2018 15:45

Group Number: 2010206

LCS/LCSD (continued)

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Perfluorononanesulfonate(Pfns)	5.22	5.01	5.22	5.27	96	101	66-133	5	30
Pfba-Perfluorobutanoic Acid	5.44	5.34	5.44	5.56	98	102	74-142	4	30
Pfbs-Perfluorobutanesulfonate	4.81	4.77	4.81	4.84	99	101	73-128	1	30
Pfda-Perfluorodecanoic Acid	5.44	5.46	5.44	5.76	100	106	69-148	5	30
Pfdoda-Perfluorododecanoic	5.44	6.14	5.44	5.43	113	100	75-136	12	30
Pfds-Perfluorodecanesulfonate	5.24	4.86	5.24	4.99	93	95	60-135	3	30
Pfhpa-Perfluoroheptanoic Acid	5.44	5.38	5.44	5.13	99	94	76-140	5	30
Pfhps-Perfluoroheptanesulfonat	5.18	4.90	5.18	4.97	95	96	64-135	1	30
Pfhxa-Perfluorohexanoic Acid	5.44	5.40	5.44	5.01	99	92	75-135	8	30
Pfhxs-Perfluorohexanesulfonate	5.14	4.91	5.14	5.12	95	100	71-131	4	30
Pfna-Perfluorononanoic Acid	5.44	4.84	5.44	4.94	89	91	72-148	2	30
Pfoa-Perfluorooctanoic Acid	5.44	5.75	5.44	5.58	106	103	72-138	3	30
Pfosa-Perfluorooctanesulfonami	5.44	4.60	5.44	4.65	85	85	65-164	1	30
Pfos-Perfluorooctanesulfonate	5.20	4.54	5.20	5.08	87	98	67-138	11	30
Pfpea-Perfluoropentanoic Acid	5.44	5.39	5.44	4.83	99	89	74-134	11	30
Pfpes-Perfluoropentanesulfonat	5.10	5.12	5.10	5.09	100	100	76-127	1	30
Pfteda-Perfluorotetradecanoic	5.44	5.48	5.44	5.34	101	98	74-135	2	30
Pftrda-Perfluorotridecanoic Ac	5.44	5.68	5.44	5.63	104	104	61-145	1	30
Pfunda-Perfluoroundecanoic Aci	5.44	4.44	5.44	5.34	82	98	75-146	18	30

Labeled Isotope Quality Control

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 18321001

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
9903347	93	95	84	98	92	94
9903348	62	68	53	69	26*	79
9903349	90	81	81	105	90	89
9903350	98	87	88	129	99	97
Blank	112	113	103	118	110	110
LCS	89	88	78	99	84	79
LCSD	98	100	89	102	94	88
Limits:	33-123	31-157	26-148	21-182	35-138	34-126
	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
9903347	93	118	96	90	98	90
9903348	78	374*	82	77	98	80
9903349	90	145	91	90	93	89

*- Outside of specification

- (1) The result for one or both determinations was less than five times the LOQ.
- (2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica
Reported: 11/29/2018 15:45

Group Number: 2010206

Labeled Isotope Quality Control (continued)

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 18321001

	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
9903350	103	138	99	95	106	90
Blank	114	149	113	115	120	111
LCS	89	104	89	90	95	85
LCSD	98	125	99	98	104	97
Limits:	35-126	32-170	48-122	50-121	41-144	47-125
	13C2-8:2-FTS	d3-NMeFOSAA	13C7-PFUnDA	d5-NeFOSAA	13C2-PFDoDA	13C2-PFTeDA
9903347	99	109	90	89	88	94
9903348	298*	145*	117	137	135*	153*
9903349	113	112	92	97	100	97
9903350	109	99	95	95	104	96
Blank	126	152*	110	111	104	106
LCS	99	116	89	90	82	80
LCSD	106	108	92	99	92	93
Limits:	27-164	30-127	30-128	30-142	39-130	26-119
	13C8-PFOSA					
9903347	99					
9903348	77					
9903349	95					
9903350	94					
Blank	111					
LCS	87					
LCSD	95					
Limits:	11-127					

*- Outside of specification

- (1) The result for one or both determinations was less than five times the LOQ.
- (2) The unspiked result was more than four times the spike added.

Ver: 09-30-2016 12/10/2018



Lancaster Laboratories
Environmental

Sample Administration Receipt Documentation Log

Doc Log ID: 233532



Group Number(s): 2010206

Client: Test America

Delivery and Receipt Information

Delivery Method:	<u>Fed Ex</u>	Arrival Timestamp:	<u>11/16/2018 11:40</u>
Number of Packages:	<u>2</u>	Number of Projects:	<u>9</u>

Arrival Condition Summary

Shipping Container Sealed:	Yes	Sample IDs on COC match Containers:	Yes
Custody Seal Present:	Yes	Sample Date/Times match COC:	Yes
Custody Seal Intact:	Yes	VOA Vial Headspace $\geq$ 6mm:	N/A
Samples Chilled:	Yes	Total Trip Blank Qty:	0
Paperwork Enclosed:	Yes	Air Quality Samples Present:	No
Samples Intact:	Yes		
Missing Samples:	No		
Extra Samples:	No		
Discrepancy in Container Qty on COC:	No		

Unpacked by Ariel Garcia (15332) at 13:22 on 11/16/2018

Samples Chilled Details

Thermometer Types: DT = Digital (Temp. Bottle) IR = Infrared (Surface Temp) All Temperatures in °C.

Cooler #	Thermometer ID	Corrected Temp	Therm. Type	Ice Type	Ice Present?	Ice Container	Elevated Temp?
1	DT42-03	0.5	DT	Wet	Y	Loose	N
2	DT42-03	0.6	DT	Wet	Y	Loose	N

Explanation of Symbols and Abbreviations

The following defines common symbols and abbreviations used in reporting technical data:

BMQL	Below Minimum Quantitation Level	mL	milliliter(s)
C	degrees Celsius	MPN	Most Probable Number
cfu	colony forming units	N.D.	non-detect
CP Units	cobalt-chloroplatinate units	ng	nanogram(s)
F	degrees Fahrenheit	NTU	nephelometric turbidity units
g	gram(s)	pg/L	picogram/liter
IU	International Units	RL	Reporting Limit
kg	kilogram(s)	TNTC	Too Numerous To Count
L	liter(s)	µg	microgram(s)
lb.	pound(s)	µL	microliter(s)
m3	cubic meter(s)	umhos/cm	micromhos/cm
meq	milliequivalents	MCL	Maximum Contamination Limit
mg	milligram(s)		
<	less than		
>	greater than		
ppm	parts per million - One ppm is equivalent to one milligram per kilogram (mg/kg) or one gram per million grams. For aqueous liquids, ppm is usually taken to be equivalent to milligrams per liter (mg/l), because one liter of water has a weight very close to a kilogram. For gases or vapors, one ppm is equivalent to one microliter per liter of gas.		
ppb	parts per billion		
Dry weight basis	Results printed under this heading have been adjusted for moisture content. This increases the analyte weight concentration to approximate the value present in a similar sample without moisture. All other results are reported on an as-received basis.		

Analytical test results meet all requirements of the associated regulatory program (i.e., NELAC (TNI), DoD, and ISO 17025) unless otherwise noted under the individual analysis.

Measurement uncertainty values, as applicable, are available upon request.

Tests results relate only to the sample tested. Clients should be aware that a critical step in a chemical or microbiological analysis is the collection of the sample. Unless the sample analyzed is truly representative of the bulk of material involved, the test results will be meaningless. If you have questions regarding the proper techniques of collecting samples, please contact us. We cannot be held responsible for sample integrity, however, unless sampling has been performed by a member of our staff.

This report shall not be reproduced except in full, without the written approval of the laboratory.

Times are local to the area of activity. Parameters listed in the 40 CFR Part 136 Table II as "analyze immediately" are not performed within 15 minutes.

WARRANTY AND LIMITS OF LIABILITY - In accepting analytical work, we warrant the accuracy of test results for the sample as submitted. THE FOREGOING EXPRESS WARRANTY IS EXCLUSIVE AND IS GIVEN IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED. WE DISCLAIM ANY OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING A WARRANTY OF FITNESS FOR PARTICULAR PURPOSE AND WARRANTY OF MERCHANTABILITY. IN NO EVENT SHALL EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL, LLC BE LIABLE FOR INDIRECT, SPECIAL, CONSEQUENTIAL, OR INCIDENTAL DAMAGES INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOSS OF PROFIT OR GOODWILL REGARDLESS OF (A) THE NEGLIGENCE (EITHER SOLE OR CONCURRENT) OF EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL AND (B) WHETHER EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL HAS BEEN INFORMED OF THE POSSIBILITY OF SUCH DAMAGES. We accept no legal responsibility for the purposes for which the client uses the test results. No purchase order or other order for work shall be accepted by Eurofins Lancaster Laboratories Environmental which includes any conditions that vary from the Standard Terms and Conditions, and Eurofins Lancaster Laboratories Environmental hereby objects to any conflicting terms contained in any acceptance or order submitted by client.

Data Qualifiers

Qualifier	Definition
C	Result confirmed by reanalysis
D1	Indicates for dual column analyses that the result is reported from column 1
D2	Indicates for dual column analyses that the result is reported from column 2
E	Concentration exceeds the calibration range
K1	Initial Calibration Blank is above the QC limit and the sample result is ND
K2	Continuing Calibration Blank is above the QC limit and the sample result is ND
K3	Initial Calibration Verification is above the QC limit and the sample result is ND
K4	Continuing Calibration Verification is above the QC limit and the sample result is ND
J (or G, I, X)	Estimated value $\geq$ the Method Detection Limit (MDL or DL) and $<$ the Limit of Quantitation (LOQ or RL)
P	Concentration difference between the primary and confirmation column $>40\%$. The lower result is reported.
P^	Concentration difference between the primary and confirmation column $>40\%$. The higher result is reported.
U	Analyte was not detected at the value indicated
V	Concentration difference between the primary and confirmation column $>100\%$. The reporting limit is raised due to this disparity and evident interference.
W	The dissolved oxygen uptake for the unseeded blank is greater than 0.20 mg/L.
Z	Laboratory Defined - see analysis report

Additional Organic and Inorganic CLP qualifiers may be used with Form 1 reports as defined by the CLP methods. Qualifiers specific to Dioxin/Furans and PCB Congeners are detailed on the individual Analysis Report.

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: IPP - Avon

TestAmerica Job ID: 190-18036-1

Laboratory: TestAmerica Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Michigan	State Program	5	57	05-05-20

Chain of Custody Record

Client Information Client Contact: Cindy Tomaszewski City of Cadillac Utilities Address: 1121 Plett Road City: Cadillac State, Zip: MI, 49601 Phone: 231-775-2368 Email: lab@cadillac-mi.net Project Name: City of Cadillac Site: 1PP-AVON		Sample ID: CINDY A. TOMASZEWSKI Phone: 231-775-2368 E-Mail: sue.schafer@testamericainc.com		Carrier Tracking No(s): COC No: 190-21738-914.1 Page: Page 1 of 1 Job #:	
Due Date Requested: TAT Requested (days): STANDARD PO #: 2019-040 WO #: TBD Project #: 19001884 SSOW #:		Analysis Requested			
Sample Identification FIELD BLANK AVON AUTO AVON 5TH STREET AVON DUPLIC.  190-18036 Chain of Custody		Sample Date 11-7-18 11-7-18 11-7-18 11-7-18		Sample Time 1110 1115 1125 —	
Sample Type (C=Comp, G=grab) G G G G		Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air) Water Water Water Water		Preservation Code: N N N N	
Field Filtered Sample (Yes or No) N N N N		Perform MS/MSD (Yes or No) N N N N		PFC, IDA - PFA, Standard List (24 Analytes) N N N N	
Total Number of containers 2 2 2 2		Special Instructions/Note: No equip: blank collected. 6/17		Preservation Codes: A - HCL B - NaOH C - Zn Acetate D - Nitric Acid E - NaHSO4 F - MeOH G - Anchor H - Ascorbic Acid I - Ice J - DI Water K - EDTA L - EDA Other: M - Hexane N - None O - AsNaO2 P - Na2O4S Q - Na2SO3 R - Na2SO3 S - H2SO4 T - TSP Dodecahydrate U - Acetone V - MCAA W - pH 4-5 Z - other (specify)	
Possible Hazard Identification ☐ Non-Hazard ☐ Flammable ☐ Skin Irritant ☐ Poison B ☐ Unknown ☐ Radiological					
Deliverable Requested: ☒ I, II, IV, Other (specify)					
Empty Kit Relinquished by:					
Relinquished by: UPS Date/Time: 11-7-18/1400 Company: UPS					
Relinquished by: Ship via UPS Ground Date/Time: 11/8/18 12:41 Company: UPS					
Relinquished by:					
Custody Seals Intact: ☒ Yes ☐ No Custody Seal No.:					
Cooler Temperature(s) °C and Other Remarks:					

☐ MSDS or Known Hazard Information Supplied by Client
☐ Bottle stickers applied ☐ ELEMENT comment entered ☐ MSDS COC scanned emailed to EH-SS
☐ Discrepancies Client ID City of Cadillac Utilities
☐ Short Hold Work Order # 190-18036
☐ Rush ☐ 24hr ☐ 2day ☐ 3day ☐ 5day ☐ Other
 Receipt evaluation performed by - Initials AmY Date 11/8/18 Time 12:41

Cooler/Sample Receipt

(AFTER HOURS) Receive complete gray areas
 Place cooler in walk-in place in this form in Receiving area
 Date Time Recd Initials

Method of Shipment:

☐ Walk-In Client ☐ TestAmerica Field/Courier
☐ Other Client/3rd Party Courier
☐ Fed Ex Tracking #
☒ UPS Tracking # 1ZE47491035439 2526
☐ Other Ground

Shipping Container Type:

☒ Cooler ☐ Box
☐ None ☐ Other
 Packing Materials:
☐ Plastic Bags ☐ Foam
☒ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☒ Other Air packs

Custody Seals Intact:

☒ Yes 615624/25 ☐ No
☐ N/A (not used or required)
 Cooling Materials:
☒ Ice (solid) ☐ Ice (Melted)
☐ Blue Ice ☐ None
☐ Other

Bacteriological Temp (°C) Corrected
 Samples

Frozen
 yes no

Received within 2 hours
 yes no

Sample Flagged
 yes no

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Received on same day	Acceptable?	Cooler ID	Note Affected Samples if temperature not acceptable
140252483			☐	☐	☐ Yes ☐ No	☐ Yes ☐ No		
140252476			☐	☐	☐ Yes ☐ No	☐ Yes ☐ No		
CP313207	1.9	1.2	☐	☒	☐ Yes ☐ No	☒ Yes ☐ No		

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 Sample Receipt and Login, Attachment 2 - Holding Times, Preservation and Container Requirements

Receipt Questions**	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	☒			
Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	☒			
Appropriate containers used & adequate volume provided?	☒			Preserved Bottles Checked with pH Strips* Yes No
Number of sample containers match COC?	☒			
Samples received within hold time?	☒			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	☒			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			

** May not be applicable if samples are not for compliance testing

* Excludes FOG, Volatiles, TOC Vials

Client Contact Record

Contact via: ☐ Phone ☐ Email ☐ Other Person Contacted: Date/Time:
☐ Discrepancy allowance agreement is on record in the client project file
 Discussion/Resolution:

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory

Reviewed by PM Signature Date

WI Page 1 of 1

WI No DT-SCA-WI-001 10
 effective 06/11/12

TestAmerica Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING



Client Information (Sub Contract Lab) Client Contact: _____ Shipping/Receiving: _____ Company: TestAmerica Laboratories, Inc. Address: 880 Riverside Parkway, West Sacramento, CA 95605 Phone: 916-373-5600 (Tel) 916-372-1059 (Fax) Email: _____ Project Name: IPP - Avon Site: _____		Sampler: _____ Lab P/N: Schaefer, Sue E-Mail: sue.schaefer@testamericainc.com State of Origin: Michigan Carrier Tracking No(s): _____ COC No: 190-21045.1 Page: 1 of 1 Job #: 190-18036-1	
Analysis Requested Due Date Requested: 12/4/2018 TAT Requested (days): _____ PO #: _____ WO #: _____ Project #: 19001884 SSOW #: _____		Preservation Codes: A - HCL B - NaOH C - Zn Acetate D - Nitric Acid E - NaHSO4 F - MeOH G - Amchlor H - Ascorbic Acid I - Ice J - DI Water K - EDTA L - EDA Z - other (Specify) _____ M - Hexane N - None O - AsH2O2 P - Na2OAS Q - Na2SO3 R - Na2S2O3 S - H2SO4 T - TSP Dodecahydrate U - Acetone V - NCA4 W - pH 4-5 Other: _____	
Sample Identification - Client ID (Lab ID) Field Blank (190-18036-1) Avon Auto (190-18036-2) Avon 5th Street (190-18036-3) Avon Duplic. (190-18036-4)		Analysis Requested PFC, IDA/3535, PFC PFAS, Standard List (24 Analytes) Perform MS/MSD (Yes or No) Field Filtered Sample (Yes or No) Preservation Code: _____ Matrix (W=water, S=solid, O=overseal, BT=Basin, A=Air) Sample Type (C=Comp, G=grab) Sample Time Sample Date Total Number of Containers	
Special Instructions/Note: _____		Special Instructions/Note: _____	
Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/test/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the Signed Chain of Custody attesting to said compliance to TestAmerica Laboratories, Inc.			
Possible Hazard Identification Unconfirmed Deliverable Requested: I, II, III, IV, Other (specify) _____ Primary Deliverable Rank: 2			
Sample Disposal (A fee may be assessed if samples are retained longer than 1 month) ☐ Return To Client ☐ Disposal By Lab ☐ Archive For _____ Months Special Instructions/QC Requirements: _____			
Empty Kit Relinquished by: _____ Relinquished by: _____ Relinquished by: _____ Relinquished by: _____		Date: _____ Date/Time: 11/9/18 1535 Date/Time: _____ Date/Time: _____	
Company: _____ Received by: _____ Received by: _____ Received by: _____		Date/Time: 11-9-18 925 Date/Time: _____ Date/Time: _____	
Custody Seal No.: 10110130 Custody Seals Intact: ☒ Yes ☐ No		Cooler Temperature(s) °C and Other Remarks: 4.5	